

MOI GIRLS' ELDORET 2025 PREDICTION



POSSIBLE KCSE EXAM COMPUTER STUDIES

451/1

PAPER 1

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTION TO CANDIDATES

- ❖ Write your name and index number in the spaces provided above
- ❖ *This paper consists of **two** sections **A** and **B**.*
- ❖ *Answer **ALL** questions in section **A**.*
- ❖ *Answer question **16** and any other **THREE** questions from section **B**.*
- ❖ *All answers should be written in the spaces provided on the question paper.*

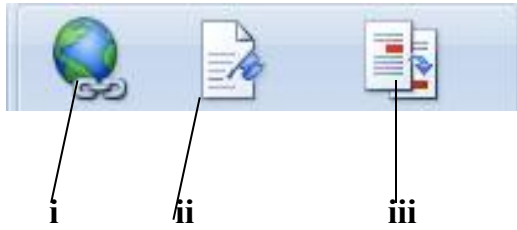
For Examiners Use Only

Section	Questions	Candidates Score
A	1-15	
B	16	
	17	
	18	
	19	
	20	
	TOTAL SCORE	

QUESTIONS

1. Name the parts labeled in the diagram below of a page layout properties

[3 marks]



i- _____

ii- _____

iii- _____

2. Explain briefly how an uninterruptible power supply (UPS) may help to protect a computer system and its data.

[1 mark]

3. How is data verification different from data validation?

[1 mark]

Give one example of data validation technique

[1 mark]

Give one example of data verification technique

[1 mark]

4. What kind of program is normally contained in a ROM chip and Explain briefly the main purpose of such a program.

[2 marks]

5. Distinguish between system software and application software. **[2 mark]**

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6. Give the similarities between optical mark readers and optical character readers. **[2 marks]**

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7. In word processing, distinguish between text insertion and overtyping. **[2 marks]**

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8. Explain, using examples where appropriate, the meaning of these computer terms.

(i) interrupt **[2 Marks]**

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(ii) buffer **[2 Marks]**

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9. Explain the following spreadsheet concepts:

(i) Automatic Recalculation. **[1 mark]**

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(ii)What if analysis

[1 mark]

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.....
10. Explain briefly why the advancement of computer technology has caused environmental hazards.

[2 marks]

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.....
.....
(i)What is telecommuting?

[1 mark]

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.....
(ii)State **four** benefits of telecommuting.

[2 marks]

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.....
.....
(ii)How is e-commerce different from e-business?

[1 mark]

.....
11. (i)What is a coprocessor?

[1 mark]

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.....
(ii)How does parallel processing work?

[1 mark]

.....

(iii)What is pipelining? [1 mark]

.....

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12. (i)What is a driver program? [1 mark]

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13. (ii). Give **one** example of hardware device that may serve more than one purpose (e.g., input and output). [2 marks]

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14. (i)How is a patent different from a trademark? [2 marks]

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(ii)What is the difference between unauthorized access and unauthorized use? [2 marks]

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15. (i)What is a workstation? [$\frac{1}{2}$ mark]

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(ii)What is a server? [$\frac{1}{2}$ mark]

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SECTION B

Answer question 16 (compulsory) and any other three questions from this section

16. (a) Explain the three control structures in programming [3 Marks]
(b) State and describe three types of error can occur in programming [3 Marks]
(c)(i) State **two** differences between a compiler and an interpreter [2 marks]
(ii) Give **one** advantage of compilers over interpreters [1 mark]
(iii) Give **one** advantage of interpreters over compilers [1 mark]
(d) Draw the corresponding flowchart for the pseudo code below. [4 marks]

Assign 0 to A,

Repeat adding 1 to A,

Until A is greater than or equal to 5,

Print A.

17. (a) How is peer-to-peer network different from client/server network, considering how programs, data and information are stored? [2 marks]

- (b) (i) Give **three** advantages of optical fibers over twisted-pair wires and coaxial cables.

[3 marks]

- (ii) Suggest **two** disadvantages of optical fibers.

[1 mark]

- (c) (i) Compare asynchronous and synchronous transmission.

[1 mark]

asynchronous	synchronous

- (ii) Give one disadvantage of the c(i) above

[1 mark]

asynchronous	synchronous

- (d) (i) Compare the following Ring ,star and bus network topologies.

[1 ½ marks]

Bus NetWork	Ring NetWork	Star Network

(ii) Give one **disadvantage** of each [1 ½ marks]

Bus NetWork	Ring NetWork	Star Network

(iii) Draw a diagram to represent the ring, star and bus topologies [1 ½ marks]

Bus NetWork	Ring NetWork	Star Network

(e) (i) Explain the term upload [½ mark]

(ii) If a 56K modem is operating at full speed, how long does it take to download a 1-MB file?
[2 marks]

18. (a). Describe the role of the systems analyst. [2 marks]

(b). Describe 2 techniques used by the systems analyst in requirements specification. [2 marks]

(b). In the software development process, what happens at the analysis stage? [1 Mark]

(c) Describe fully what is meant by the term *top-down design*? [3 Marks]

(d) The process of developing such software is an iterative process. What does *iterative* mean?
[1 Mark]

(e)(i) What are CASE tools?
[1 Mark]

(ii) What is the purpose of CASE tools?
[1 Mark]

(iii) What does *CASE* stand for?

[1 Mark]

(f) Identify ant two activities within the design stage of the software development process

[3 Marks]

19 (a).Describe four functions of a single user operating system.

[4 Marks]

(b) Explain the need for an operating system to have a command language interpreter. [1 Mark]

(c) Give 4 examples of the many processes which can be initiated by a user and which are then carried out by the operating system.

[4 Marks]

(d) What happens if a command entered by the user is not recognized by the operating system?

[1 Mark]

(e)Describe an advantage and a disadvantage to storing the operating system in RAM rather than in ROM?

[2 Marks]

(g) Describe three features that a **networkoperatingsystem** must have that a single user operating system does not need.

[3 Marks]

20. (a) Explain two reasons why computers use the binary system?

[2 Marks]

(b)What is meant by the term *word* in the phrase *16-bit word computer*?

[1 Mark]

(c) What is the decimal value of the largest integer that can be represented by a byte in a computer which deals only with positive numbers?

[1 Mark]

(d)What does the acronym ASCII stand for and what is ASCII?

[2 Marks]

(e) Complete the following table and note that no marks will be awarded for answers with missing or unclear working.

[6 Marks]

BIN	DEC	HEX
	45	
		B6
11001		